

TIANREN CHEN

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EDUCATION

University of California, Irvine
M.S. in Computer Science (CS)

Expected Graduation: December 2027
Irvine, CA

University of California, Davis
B.S. in Computer Science (CS)

June 2025
Davis, CA

RELEVANT SKILLS

- **Languages:** Python, C/C++, Java, Assembly, SQL, JavaScript, HTML/CSS
- **Frameworks/Tools:** Flask, Node.js, Express, React, MySQL, SQLite, Docker, Git/GitHub
- **Areas:** Backend Development, Full-Stack Development, Machine Learning, Distributed Systems

WORK EXPERIENCE

Megawatt, Data Center Network Deployment Intern

June 2026 – August 2026

- Validated and troubleshoot server-to-leaf and leaf-to-spine connectivity by analyzing port status, endpoint mappings, Tx/Rx polarity, link speed, and physical-layer failures.
- Diagnosed fiber and InfiniBand link failures using BER testing, optical diagnostics, cable tracing, and port verification.

American Eagle Immigration Law Firm, Software Developer

July 2025 – December 2025

- Developed a Python and Flask-based Auto Form application to extract, transform, validate, and generate structured immigration application data.
- Implemented backend data-processing logic including field-level validation, schema validation, data normalization, exception handling, and missing-data detection.
- Automated repetitive document-processing workflows by integrating structured client data with form-generation logic, reducing manual data-entry requirements.

Golden Fastrack Corp, Summer Intern

June 2023 – September 2023

- Wrote SQL queries to retrieve, update, and maintain business records across internal databases.
- Performed data integrity checks and troubleshoot duplicate, missing, and inconsistent records.

PROJECTS

Distributed File System Project

- Developed a multithreaded distributed file system in C++ supporting concurrent file operations across multiple clients.
- Implemented RESTful APIs for GET, PUT, and DELETE operations and designed request-handling logic for file access and modification.
- Applied mutexes and condition variables to coordinate concurrent requests and prevent race conditions during shared-resource access.
- Implemented FIFO scheduling and transactional logic to maintain consistency across concurrent file operations.

Full-Stack Web Application

- Built a full-stack web application using Node.js, Express, SQLite, and Handlebars with persistent user and application data.
- Integrated Google OAuth 2.0 authentication with Passport.js to support secure user login and session management.
- Developed backend API endpoints and database operations for retrieving, creating, and updating application data.
- Integrated third-party APIs using asynchronous HTTP requests and dynamically rendered retrieved data in the web interface.

Movie Rating Prediction System

- Developed an end-to-end machine learning pipeline to predict IMDb movie ratings using data integrated from multiple public datasets.
- Implemented data preprocessing, feature engineering, model training, and evaluation workflows in Python.
- Compared Random Forest, XGBoost, and neural network models and tuned model configurations to improve predictive performance.